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Child health care & Growth Monitoring

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Objectives

- To explain the child health care and preventive services.
- To understand the goals of child health care.
- To identify parts of child health care program.
- To describe domains of child development
- To explain growth monitoring of children under 5 years old..
- To Interpret growth charts and identify growth abnormalities.
- To explain causes of malnutrition

Child Health Care program

Definition: Child health care refers to comprehensive health services provided to promote optimal growth and development of children under the age of 5 years old.

Preventive services are needed for children for the fallowing reasons:-

1. Children represent high percentage of population (40% -45%) .

2.Many causes of morbidity and mortality are avoidable such as :

- Accidental and non- accidental injuries poisoning , drugs , electricity , falls from bed or stairs, drowning and road traffic accidents .

inside and outside home

- Sudden Infant Death Syndrome (S I D S). → can be avoided
- Infectious disease which can be prevented by Immunization,good personal hygiene and sanitation., parent education

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Goal of Child Health Care

↑ mother-child bond
↓ child illness & death
provide nutrient

- **Promotive:** Health education, breastfeeding promotion, immunization awareness
- **Preventive:** Immunization, screening for developmental delay, nutritional supplementation
- **Curative:** Management of common childhood illnesses (ARI, diarrhea, malnutrition)

  Acute Respiratory infection
- **Rehabilitative:** Care for children with chronic illness or disability, family counseling., physical therapy

Parts of child health care program

1. Growth Monitoring
2. Control of diarrheal disease
3. Breast feeding
4. Expanded Program of Immunization
5. Family Planning
6. Food fortification and supplementation
7. Female education
8. Control of Acute Respiratory Infection (ARI).

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Growth monitoring

Growth and Development

getting bigger

physical

- **Growth:** Increase in size and weight of the body and organs (quantitative). like height, weight, head circumference

- **Development:** , Improvement in the child's abilities and functions— e.g., motor, language, and cognitive abilities, (qualitative)

getting better

Domains of development

large movement (muscles)

- Gross Motor: Sitting, walking, running

small movement (muscles)

- Fine Motor: Grasping, drawing
- Language: Babbling, using words, and forming sentences.
- Social/Emotional: Smiling, playing, sharing
- Cognitive: Problem-solving, attention, and understanding new information.

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certain skills and abilities that children usually do at a specific age

Developmental of Milestones

- **Age 2 months:** Head control, follows objects, smilling
- **6 months:** Sits with support, transfers objects, babbles, recognizes faces
- **9 months**: Crawls, says 'mama/dada', stranger anxiety
- **12 months:** Walks with support,using words, waves bye-bye

Growth Monitoring

started at birth
- measuring height, weight, head circumference
- APGAR
- 2 days later detailed pediatric examination
- vaccine program started

Growth monitoring is a process of sequential measurements of the weight of children under the age of five years , in order to be able to detect signs of malnutrition , (growth failure) as early as possible and correct this failure.

For every new born baby 2 routine examination will be carried:

1. Estimation of Apgar score:

The Apgar score provides an accepted and convenient method for reporting the status of the newborn infant immediately after birth and the response to resuscitation if needed.

APGAR Evaluation of newborn infants

	SIGN	0	1	2
Pulse (P)	Heart rate	Absent	Below 100	Over 100
Respiration (R)	Respiratory effort	Absent	Slow, irregular	Good, crying
Activity (A)	Muscle tone	Limp	Some flexion	Active motion
Grimace (G)	Reflex*	No response	Grimace	Cough or sneeze
Appearance (A)	Color	Blue, Pale	Body pink, Extremities blue	Completely pink

- 7 to 10 is normal
- 4 to 6 is moderately depressed
- 0 to 3 needs immediate resuscitation

getting 10 is highly rare

*Response to catheter in nostrils

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2. Late new-born care:

- The second part of this examination should be done within the first 48 hours after birth by a pediatrician. much more detailed than APGAR

Growth Monitoring and promotion

- Growth monitoring and promotion can be defined as an operational strategy enabling the mothers to visualize growth or lack of growth of her child.
- Growth monitoring can also be referred to as the periodic weighing of children and the plotting of each measurement on a growth chart or child health card.

- **Use the Growth Record:**

Select a Boy's Growth Record or Girl's Growth Record.

1.This booklet will be record of the child's growth and health.

2. In each time visiting, the child will be weighed and measured, and the measurements will be recorded in this booklet.

3. The booklet includes charts on which we will plot the child's measurements in order to assess his or her growth.

Growth Chart:

- A growth chart an assessment tool used by pediatricians, family physicians, and other health care providers to follow a child growth over time.
- The height, weight, and head circumference of a child and compare it to standard growth patterns

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Two types of growth chart:

used in regulary distributed data → here data has mean (average) → which Z-score use mean (SD)

- **Z score chart :** The Z score is SD above or below the mean. It is a statistical measurement that tell us how measurement far from mean.

use percentage → for non-regular distributed data → here we have median → percentile use median

- **Percentile chart:** The chart uses percentiles to show how a child compares to others of the same age and sex.
- A Z score of 0 is at the (ideal) apex of the curve and it is the same as a 50th percentile , Z score of±1.0 plot at the 15th or 85th percentile respectively, Z score of±2.0 plot at the 3rd or 97th percentile respectively.

The measurement should be taken and recorded whenever an infant or child visit a health care provider for an immunization, check-up, or care during illness.

By measuring height and weight we obtain the following categories :

1. Weight for Age Z score (WAZ)

- -1 < WAZ < 0 Normal (well nourished).
- -2 < WAZ < -1 Mildly malnourished
- -3 < WAZ < -2 Moderately malnourished
- -4 < WAZ < -3 severely malnourished.
- In the case of having WAZ below mean, it is an indication of malnutrition

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2. Height for Age Z score (WAZ)

- -1 < HAZ < 0 Normal → 1<2 in here are also normal, +3 is tall usually due to having tall parent
- -2 < HAZ < -1 Mildly Stunted
- -3 < HAZ < -2 Moderately stunted
- -4 < HAZ < -3 severely Stunted.

In the case of having HAZ below mean is called Stunting and is an indication of malnutrition.

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3. Weight for Height Z score (WHZ)

→ muscle wasting

- -1 < WHZ < 0 Normal
- -2 < WHZ < -1 Mildly Wasted
- -3 < WHZ < -2 Moderately Wasted
- -4 < WHZ < -3 severely wasted.

In the case of having WHZ below mean is called Wasted and is an indication of malnutrition.

4. Other Z score:

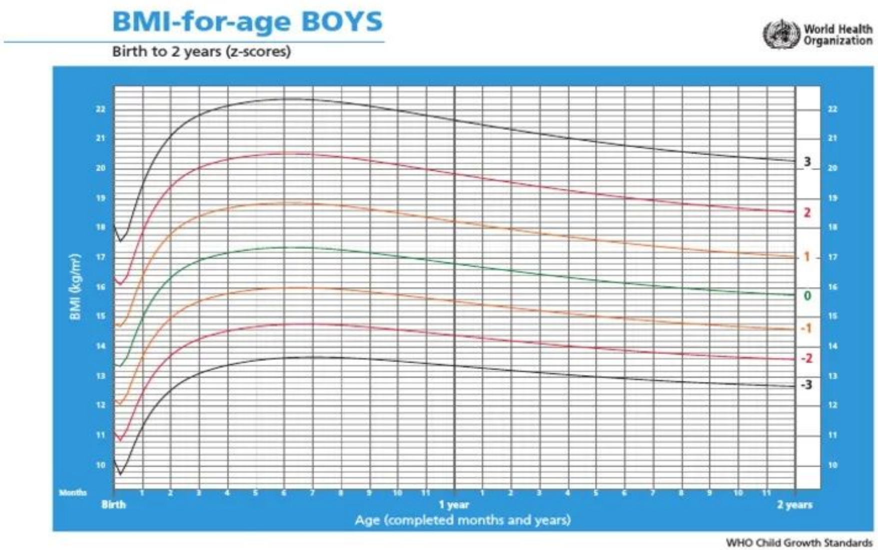
- BMI for age and HC for age Z score

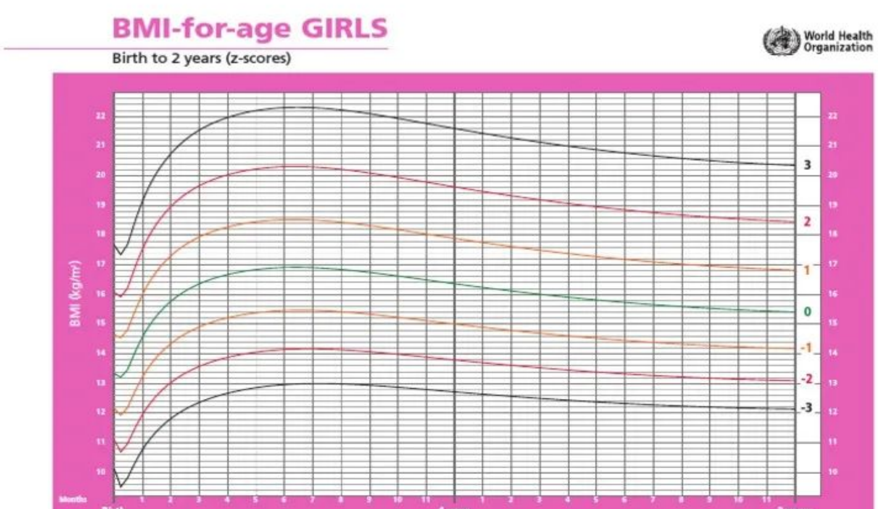
→ not as chart in our country

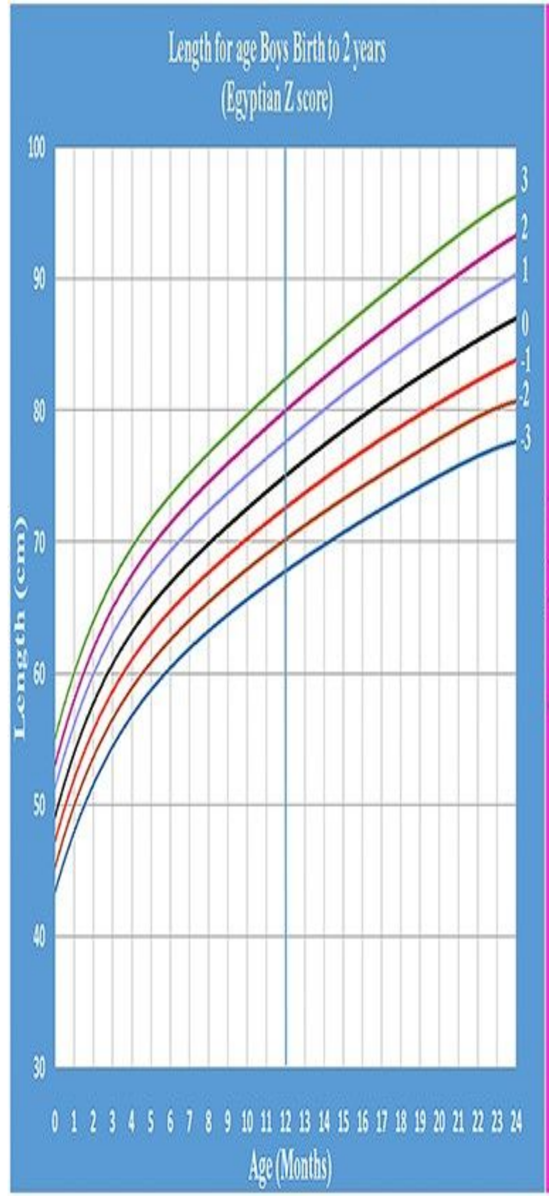
Growth Charts

in our country
0-6 month
6-2
2-5 years

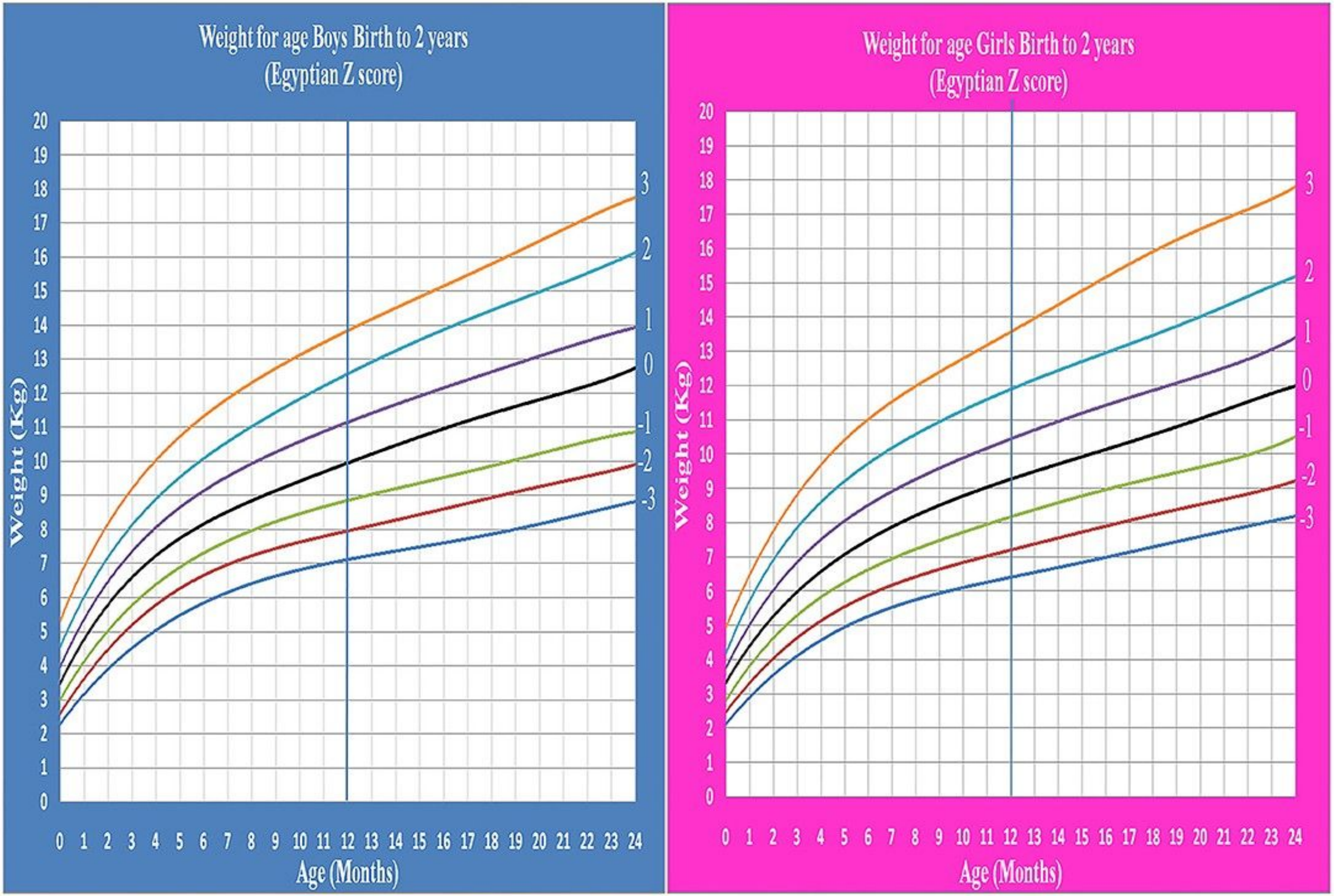
each has 4 charts → weight for Age, height for Age, weight for height, BMI



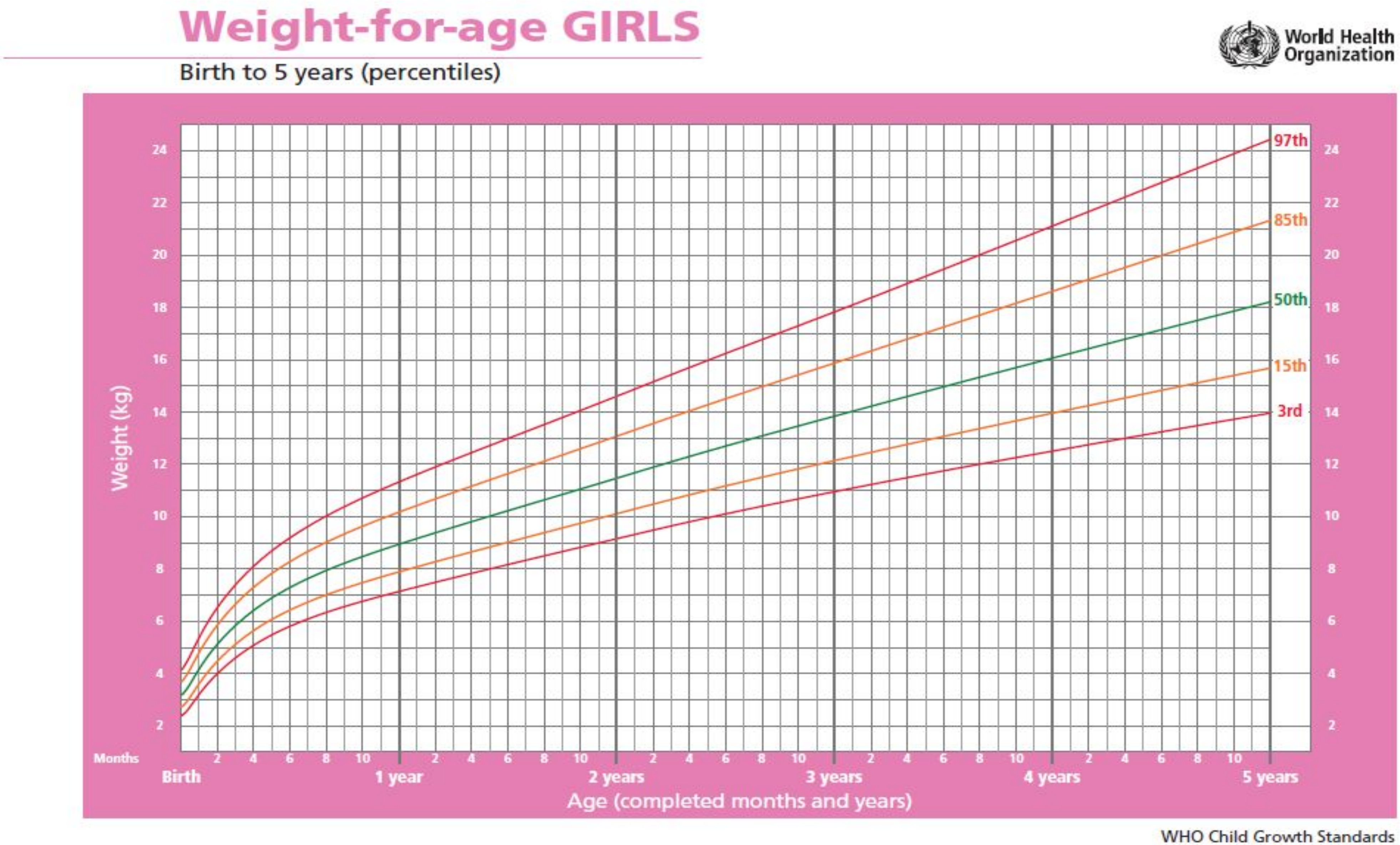




Growth Charts



Growth chart



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Headcircumference HC:

- It is recommended that the infant's HC recorded after birth and before discharge.
- A routine measurement of head circumference is intended to aid the detection of two groups of disorders:(macrocephaly) and (microcephaly).
- A head circumference above the 97th percentile or below the 3rd percentile at any stage is an indication for more detailed assessment. HC alone is not diagnostic, associated symptoms should be present if not then more frequent follow-up

- Regular weight gain is the most important sign that a child is growing well.
- Each child should have a growth chart with a line that is showing how well the child is growing.
- If the line goes up gradually, the child is doing well. A line that stays flat or goes down indicates malnutrition.

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Malnutrition:

- Malnutrition is a condition that occurs when the body does not get the right amount of nutrient, too little or too much or imbalance needed for growth and development.
- Is very prevalent problem among under 5 years children especially in the developing world .
- Most malnutrition is invisible (Iceberg phenomenon). insidious till becomes very severe
- Only 1-2% of all malnutrition children will show signs of clinical malnutrition.

- Marasmus and kwashiorkor which are sever malnutrition

Causes of malnutrition

- Inadequate breastfeeding or poor complementary feeding

  since in simple word beginning of breast feeding is rapid since has no fat no nutrient but as progresses it becomes slower rate due to fat and nutrient but mother wont feel the infant is fed.

- Repeated infection (diarrhea, respiratory infection)
- Poor knowledge of mothers of type of feeding & incorrect feeding practices .
- Poverty and food insecurity.

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Sever Malnutrition in children:

more severe

No fat and carbohydrate

↓ protein
↓ caloric

Marasmus (non-edematous malnutrition):

-3 or even -4

In this form of severe malnutrition the child is severely wasted and has the appearance of “skin and bones” due to loss of muscle and fatty tissue. Weight-for-age and weight-for- height are likely to be very low.

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poverty related

- **Kwashiorkor (edematous malnutrition):** ↓ protein; constant caloric → since there only food is fat and carbohydrate

-3 to -4

In this form of severe under nutrition the child's

but fatty tissue normal

muscles are wasted, but the wasting may not be apparent due to generalized edema (swelling from excess fluid in the tissues). A child with kwashiorkor will usually be underweight, but the edema may mask the true weight .